# MTH2003

## UNIVERSITY OF EXETER

## FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

## MATHEMATICS AND STATISTICS

### January 2023

**Differential Equations**

**Module Leader: Prof. Vadim N. Biktashev**

**Duration: 2 HOURS**

**Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

## USEFUL FORMULAS

These can be used without proof if and where needed.

**An important boundary-value problem**

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(L) = 0$$

has solutions $\lambda_n = (n\pi/L)^2$, $\phi_n(x) = \sin(n\pi x/L)$, $n \in \{1, 2, 3, \dots\}$.

**An interesting trigonometric identity**

$$\sin^3(x) = \frac{1}{4}\left(3\sin x - \sin(3x)\right)$$

**Eigenfunctions of the Sturm-Liouville problem**

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(p(x)\frac{\mathrm{d}y}{\mathrm{d}x}\right) - q(x)y + \lambda r(x)y = 0, \qquad y(a) = 0, \quad y(b) = 0,$$

where $a < b$ and $p(x) > 0$, $r(x) > 0$ for $x \in (a, b)$, satisfy the orthogonality condition

$$\int_a^b \phi_n(x)\phi_m(x)r(x)\,\mathrm{d}x = 0 \quad \text{if} \quad \lambda_m \neq \lambda_n.$$

**"Sturm's oscillation theorem"** If the eigenvalues of the Sturm-Liouville problem are enumerated so that $\lambda_1 < \lambda_2 < \lambda_3 < \dots$, then an eigenfunction $\phi_n(x)$ has exactly $n - 1$ roots within $x \in (a, b)$.

## SECTION A

1. (a) Find all solutions $y(x)$ to the following ordinary differential equation:

$$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4x\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 8\,x^4$$

on the interval $x \in (0, \infty)$. Find also the particular solutions such that (i) $y(1) = -1$, $y(2) = 36$ and (ii) $y(1) = -1$, $y'(1) = 4$. (15)

(b) Write the system of ODEs

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -6x - 9y,$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -x - 6y$$

in the vector form $\mathrm{d}\mathbf{x}/\mathrm{d}t = \mathbf{A}\mathbf{x}$. Identify $\mathbf{x}$ and $\mathbf{A}$. Find the eigenvalues and eigenvectors of the matrix of coefficients $\mathbf{A}$. Hence write down the general solution $x(t)$, $y(t)$ of the system. Write down a fundamental matrix $\mathbf{\Psi}(t)$ of this system, satisfying the initial condition $\mathbf{\Psi}(0) = \mathbf{I}$. (15)

(c) Consider the eigenfunction problem

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0,\ x \in [0, \pi]; \qquad y(0) = 0,\ y(\pi) = 0.$$

Bring this equation into the Sturm-Liouville form and hence state the orthogonality condition for the eigenfunctions of this problem. Verify that $y_* = \mathrm{e}^{-2x}\sin(4x)$ is an eigenfunction of this problem and find the corresponding eigenvalue $\lambda_*$. State how many eigenvalues smaller than $\lambda_*$ this problem has. (10)

(d) Write down the general solution of the partial differential equation with boundary conditions:

$$\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}, \qquad u(0,t) = 0,\ u(\pi,t) = 0,$$

in the form of an infinite series of terms of the form $X(x)T(t)$ where $X(x)$ and $T(t)$ should be found. Find also the solution that satisfies the initial conditions $u(x,0) = 4\sin^3(x)$, $u_t(x,0) = 0$. (10)

[**50**]

## SECTION B

2. (a) Consider an initial-value problem for a generic first-order differential equation written in the resolved form

(Eq2.1) $$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y), \qquad y(a) = b.$$

Let $f_x = \partial f/\partial x$ and $f_y = \partial f/\partial y$, and assume that $f$ is defined and continuous in a neighbourhood of $(a, b)$. Select one true statement from the following list: For local unique solvability of the problem,

- ... it is sufficient that $f_x$ is continuous in a neighbourhood of the point $(a, b)$.
- ... it is sufficient that $f_y$ is continuous in a neighbourhood of the point $(a, b)$.
- ... it is necessary that $f_x$ is continuous in a neighbourhood of the point $(a, b)$.
- ... it is necessary that $f_y$ is continuous in a neighbourhood of the point $(a, b)$.

Note you are not required to give a proof. (4)

(b) Consider the ODE

(Eq2.2) $$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 2\left|y - x^2\right|^{1/2}.$$

Find out for what $a$ and $b$ the initial-value problem $y(a) = b$ for (Eq2.2) is guaranteed to be (i) locally solvable, and (ii) locally uniquely solvable. (8)

(c) Using the substitution $z = y - x^2$ in (Eq2.2), where $z = z(x)$ is the new unknown function, write down an ODE for $z(x)$. Find the typical solutions (depending on arbitrary constants) and the special solutions (any other solutions that are not captured by the typical solutions) of this ODE for $z(x)$. *Take care to identify the intervals in which the solutions exist and the values that the arbitrary constants may take.* Hence write down the general solution for (Eq2.2). Find also the solution satisfying $y(-1) = 2$, $y(2) = 3$. (13)

[**25**]

3. Consider the following ODE, known as the modified Bessel equation of order 1/2:

$$\text{(Eq3.1)} \qquad y'' + \frac{1}{x}y' - \left(1 + \frac{1}{4x^2}\right) y = 0, \qquad x > 0.$$

(a) Explain (i) why $x = 0$ is a singular point of this equation, and (ii) why $x = 0$ is a regular singular point of this equation. Find all values of the constant $m$ for which solutions of this ODE in the form

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+m}, \qquad a_n = \text{const}, \qquad a_0 \neq 0$$

exist. (6)

(b) Use the method of Frobenius to show that solutions of this equation that are bounded as $x \to 0$ can be written in the form

$$y = \sum_{n=0}^{\infty} a_{2n} x^{2n+\frac{1}{2}},$$

where the coefficients $a_n$ satisfy the recurrence relation

$$a_{2n+2} = \frac{a_{2n}}{(2n+2)(2n+3)}.$$

(8)

(c) Find the transformation of (Eq3.1) of the form

$$y(x) = q(x)z(x)$$

where $z(x)$ is the new unknown function, and $q(x) \neq 0$ is a fixed function you need to define, such that the ODE for $z(x)$ does not have the term with the first derivative $\mathrm{d}z/\mathrm{d}x$. Solve the resulting ODE for $z(x)$ and hence find the general solution of (Eq3.1). Show that this general solution is in agreement with the result of part (b) about the solutions bounded as $x \to 0$. (11)

[**25**]

4. (a) Motion of the membrane of a rectangular drum is described by a wave equation

$$\text{(Eq4.1)} \quad \frac{\partial^2 u}{\partial t^2} = \nabla^2 u, \qquad 0 \le x \le 1, \quad 0 \le y \le 1/2, \qquad t \ge 0,$$

with boundary conditions that $u(x, y, t) = 0$ whenever $x = 0$ or $x = 1$ or $y = 0$ or $y = 1/2$. Use the method of separation of variables, also known as the generalized Fourier method, to find the general solution of this equation, in the form of a double-infinite series of solutions in the form $u(x, y, t) = X(x)Y(y)T(t)$. (15)

(b) This rectangular drum will sound a number of irrationally related frequencies. What are the three lowest frequencies in its spectrum? Give the answers to 4 decimal places. (7)

(c) A *harmonic* is a solution $u(x, y, t)$ of (Eq4.1) with a particular temporal frequency $\omega$, that is,

$$u(x, y, t) = U(x, y)\,[A\cos(\omega t) + B\sin(\omega t)] \not\equiv 0,$$

and a *nodal line* of a harmonic is the set of points $(x, y)$ where this solution equals zero at all times, that is, $U(x, y) = 0$. Draw sketches of the nodal lines of the harmonics corresponding to the three lowest frequencies you identified in part (b). (3)

[**25**]